

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 3 – Comparative Analysis

Course Code:	CSA06	Course Title:	Design and Analysis of Algorithms
CO Assessed:	CO4: Articulation (BL4, BL5)	Assessment:	Assessment Tool 3 – Comparative Analysis
Weightage:	10% of CIA	Total Marks:	100
CO4 Statement:	Solve optimization problems using dynamic programming and greedy strategies.		
Student Name:	Manoj M	Reg. No.:	192511060
Section / Batch:	CSE	Date:	22.08.2026

Instructions to Students

- Answer the following question. Total Marks: 100.
- Analyze the given questions thoroughly before answering.
- Compare multiple aspects such as methods, results, advantages, and limitations clearly.
- Critically evaluate the strengths, weaknesses, and implications with proper justification.
- Present meaningful insights, interpretations, and well-supported conclusions from the comparisons.

Question Type	Focus Area	Questions
Comparative Analysis	Comparative analysis of Dynamic Programming and Greedy algorithms for optimization problems.	Q1

SECTION A – Comparative Analysis

Shortest Path Comparison (Dijkstra vs Bellman-Ford)100 marks

Given a graph with edges including negative weights, apply both Dijkstra's and Bellman-Ford algorithms to compute shortest paths. Compare their correctness, complexity, and limitations. Critically evaluate why Dijkstra fails in certain cases. Analyze performance differences and derive insights on algorithm selection.

Code of Conduct

I _____Manoj (192511060)___ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: _____

RUBRICS FOR CALCULATION

Comparative Analysis					
Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Selection of Studies/Parameters	20%	Selects highly relevant and diverse studies with well-defined comparison parameters	Selects relevant studies with minor gaps in parameter definition	Limited or partially relevant selection	Poor or inappropriate selection of studies
Comparison & Differentiation	25%	Clearly compares multiple aspects (methods, results, limitations) with strong differentiation	Good comparison with minor gaps	Basic comparison with limited depth	No clear comparison; mostly descriptive
Critical Evaluation	25%	Critically evaluates strengths, weaknesses, and implications with strong justification	Provides evaluation with some justification	Limited evaluation; lacks depth	No critical evaluation provided
Synthesis & Insight Generation	20%	Generates meaningful insights and conclusions from comparisons	Some insights derived from comparison	Limited or unclear insights	No insights generated
Clarity	10%	Information is clearly presented (tables/figures) with logical structure	Generally clear with minor issues	Presentation lacks clarity or structure	Poor presentation and difficult to understand

Marks Evaluation:

Criteria	Wt.	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
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Selection of Studies/Parameters	20%				
Comparison & Differentiation	25%				
Critical Evaluation	25%				
Synthesis & Insight Generation	20%				
Clarity	10%				
Total Marks (100):					

Faculty In-charge	Course Coordinator	Head of Department
Name: _____	Name: _____	Name: _____
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____

Algorithm: Shortest Path Comparison — Dijkstra vs Bellman-Ford

1. **Initialize** the graph with vertices, edges, source vertex, and edge weights. Set source distance to 0 and all others to ∞ .
2. **Apply Dijkstra's Algorithm:** repeatedly select the unvisited vertex with the smallest distance and relax its adjacent edges.
3. **Apply Bellman-Ford:** relax **all edges V-1 times** to calculate shortest distances, allowing negative edge weights.
4. **Check negative cycles** using one additional relaxation. If a distance improves, a negative-weight cycle exists.
5. **Compare results:** Dijkstra is faster but can give incorrect results with negative edges, while Bellman-Ford correctly handles negative weights.
6. **Analyze complexity:** Dijkstra with a priority queue $\rightarrow O((V+E) \log V)$; Bellman-Ford $\rightarrow O(VE)$.
7. **Select the algorithm:** use Dijkstra for graphs with **non-negative weights** and Bellman-Ford when **negative edge weights** are present.

Code:

 **SIMATS**
ENGINEERING

SIMATS Online Programming Lab
DAA PROGRAMMING

Manoj M
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1/1 answered

Your Code**Input**

stdin input

Output--- Dijkstra Algorithm Results ---
-
Vertex 0: 0
Vertex 1: 6
Vertex 2: 7
Vertex 3: 4